+PROJECT

REPORT ON

Secure Two-LAN Network with DHCP and Web Server

Submitted to

NMAM INSTITUTE OF TECHNOLOGY, NITTE

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In partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology

in

COMPUTER SCIENCE AND ENGINEERING (CYBERSECURITY)

by

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ABSTRACT

In modern network environments, security, efficient communication, and proper management are highly important. Many small organizations still use a single flat network where all devices are connected in the same broadcast domain, which can lead to security risks, traffic congestion, and poor network control. This project aims to overcome these issues by designing and simulating a Secure Two-LAN Network using Cisco Packet Tracer. The network is divided into two separate LANs: one for internal users and another for public/server access.

The objective of this project is to improve network security, organize communication, and demonstrate controlled connectivity between two LANs through a router. Static and DHCP-based IP addressing methods are used for proper device configuration. A web server is deployed in the second LAN to host an internal webpage accessible through the network.

The proposed solution includes router-based communication, network segmentation, and service hosting in a secure environment. The final outcome is a reliable, scalable, and efficient network model that enhances security while maintaining accessibility between authorized devices. This design is suitable for small office, educational, and enterprise environments where controlled communication and network separation are essential.

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CHAPTER 1

INTRODUCTION

1.1 Purpose

The purpose of this project is to design and simulate a Secure Two-LAN Network using Cisco Packet Tracer that supports efficient communication, network segmentation, and web server hosting. The project focuses on creating a secure and organized network infrastructure by dividing the network into two separate LANs: one for internal users and another for public/server access.

The goal is to demonstrate how routing between two LANs can be achieved through a router while maintaining better security and control over traffic flow. It also aims to show practical implementation of IP addressing, server hosting, and controlled communication between devices in different networks.

1.2 Scope

The network simulates the following:

• Two separate LAN networks for internal and public access
• Communication between LANs through a router
• Static and DHCP-based IP addressing
• Internal web server hosting
• Secure traffic flow using network separation
• Ping and HTTP connectivity testing
• Packet flow simulation using Cisco Packet Tracer

1.3 Overview

In this simulation:

• User1 and User2 systems are placed in LAN1 (Internal Network).
• Guest1, Guest2, and Web Server are placed in LAN2 (DMZ/Public Network).
• A Cisco 2911 Router is configured with two interfaces to connect both LANs.
• The router acts as a security gateway and controls communication between networks.

- A web server is hosted in LAN2 and provides webpage access across the network.
- Cisco Packet Tracer simulation mode is used to visualize packet movement and testing.

The design ensures network separation, better security, organized communication, and future scalability, making it suitable for small office, educational, and enterprise environments.

CHAPTER 2

REQUIREMENT SPECIFICATION

2.1 Hardware Specification

• Cisco 2911 Router

• Two Cisco 2960 Switches

• Four PCs (User1, User2, Guest1, Guest2)

• One Web Server

• Ethernet Cables (Copper Straight-through)

• Computer system with minimum 4 GB RAM for simulation

2.2 Software Specification

1.Cisco Packet Tracer v7.2 or Higher

• Used to design and simulate the complete two-LAN network topology.

• Provides CLI interface for configuring router and network devices.

• Supports simulation mode for analysing packet flow and testing communication.

2.Web Server Service (within Packet Tracer Server > HTTP)

• Used to host a custom internal web page accessible through the network.

3.Packet Tracer Internal Command Prompt

• Used to test connectivity using commands such as ping and IP verification.

4.Microsoft Word / Google Docs

• Used for preparing project documentation and report.

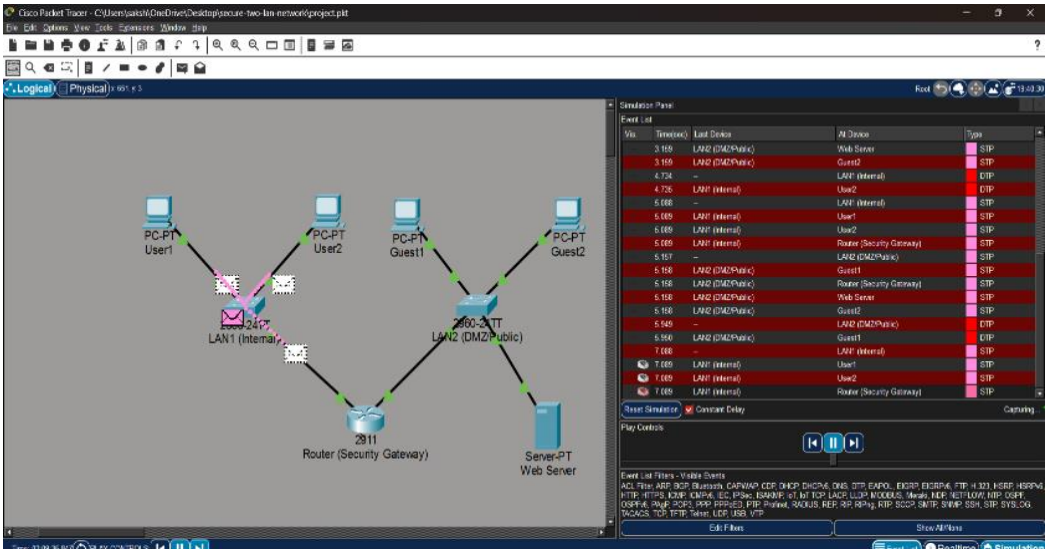

Figure 2.1: Simulation Mode in Cisco Packet Tracer

Figure 2.1 shows the Simulation Mode in Cisco Packet Tracer used to observe packet flow and analyze communication between network devices

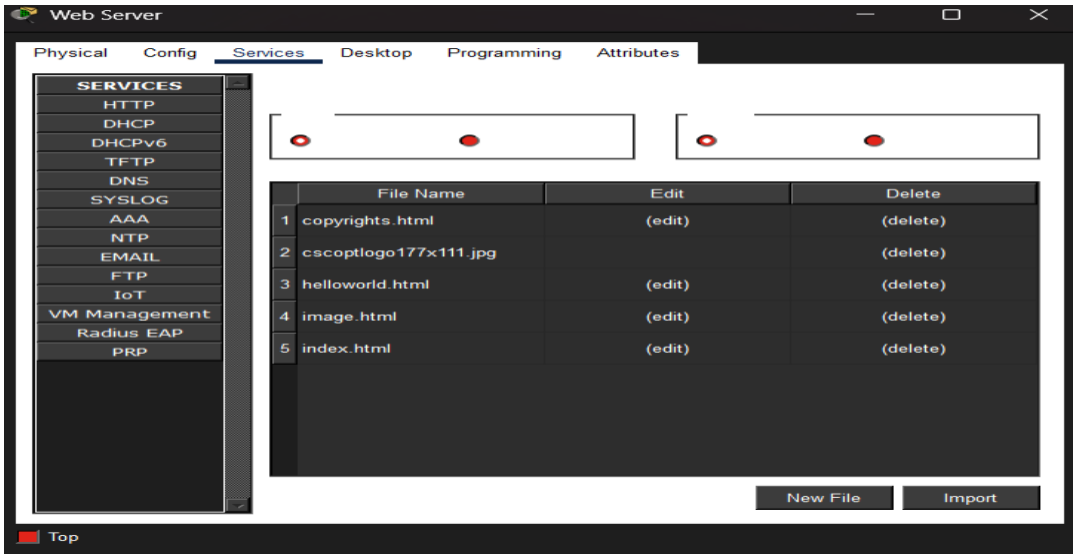

Figure 2.2: Server Services Configuration in Cisco Packet Tracer

Figure 2.2 shows the Server Services panel in Cisco Packet Tracer where the HTTP service is configured for hosting the internal web page. It confirms that the web server services are enabled and ready for network access.

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CHAPTER 3

SYSTEM DESIGN

3.1 Network Topology Diagram

The system is designed using a two-LAN network architecture in Cisco Packet Tracer. The network consists of two separate LANs connected through a router. LAN1 is used for internal users, while LAN2 is used for public users and the web server. The router acts as a security gateway and controls communication between both networks. This design provides better traffic management, network organization, and security.

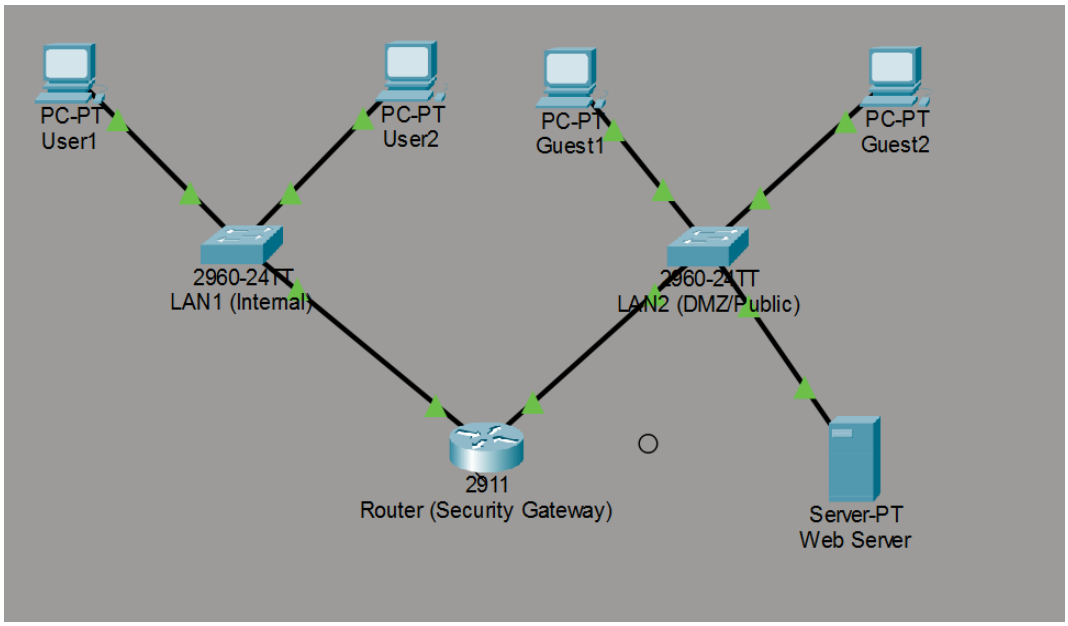

Figure 3.1: Secure Two-LAN Network Topology

Figure 3.1 illustrates the overall design of the Secure Two-LAN Network consisting of two separate LANs connected through a router. It shows the placement of user systems, switches, router, and web server for organized and secure communication

3.2 IP Addressing and LAN Planning

LAN ID	Department / Network	Subnet	Default Gateway	Devices
LAN 1	InternalNetwork	192.168.1.0/24	192.168.1.1	User1, User2
LAN	Public / DMZ	192.168.2.0/2	192.168.2.1	Guest1,Guest2

LAN ID	Department / Network	Subnet	Default Gateway	Devices
2	Network	4		, Server

Table 3.1 IP Addressing and Network Allocation Scheme

3.3 Device Role Assignment

Device	Role	Network	Description
User1, User2	Internal Users	LAN1	Used for secure internal communication
Guest1, Guest2	Public Users	LAN2	Represents external/public systems
Server	Web Server	LAN2	Hosts the internal website
Router (2911)	Security Gateway	Both LANs	Routes traffic between networks
Switch0	LAN1 Switch	LAN1	Connects internal devices
Switch1	LAN2 Switch	LAN2	Connects public devices and server

Table 3.2 Device Role Assignment Table

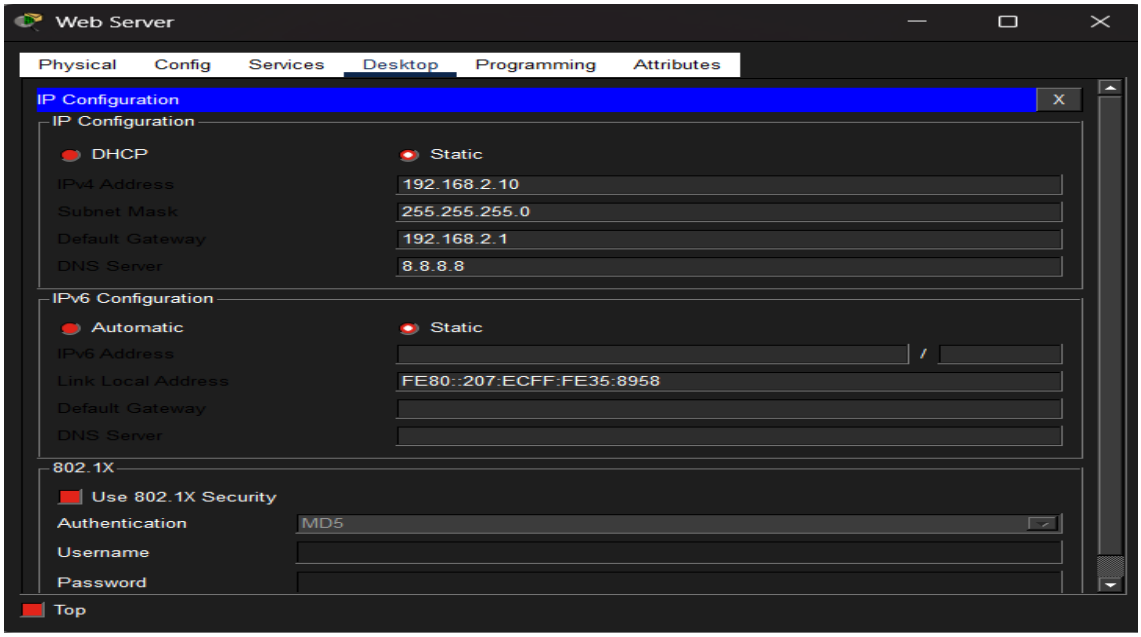

Figure 3.2: Web Server Network Address Configuration

Figure 3.2 shows the IP address configuration of the web server connected to LAN2 in the network topology. It includes the server IP address, subnet mask, and default gateway required for network communication and web hosting.

CHAPTER 4

IMPLEMENTATION

4.1 Network Topology Setup in Cisco Packet Tracer

The complete network topology was created in Cisco Packet Tracer using one Cisco 2911 router, two Cisco 2960 switches, four PCs, and one server. The devices were connected using copper straight-through cables. LAN1 was configured for internal users, while LAN2 was configured for public users and server hosting.

4.2 Router Configuration

The Cisco 2911 router was configured with two interfaces to connect both LAN networks. Each interface was assigned an IP address to act as the default gateway for the respective LAN.

• GigabitEthernet0/0 : 192.168.1.1 /24
• GigabitEthernet0/1 : 192.168.2.1 /24

The router enables communication between LAN1 and LAN2 and acts as a security gateway.

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

Figure 4.1: Router Interface Configuration

Figure 4.1 shows the configuration of router interfaces used to connect LAN1 and LAN2 networks. Each interface is assigned an IP address that acts as the default gateway for devices in the respective LAN.

4.3 PC Configuration

All PCs were configured with IP addresses according to their respective LAN networks.

LAN1 (Internal Network)

• User1 – 192.168.1.21
• User2 – 192.168.1.22

LAN2 (Public Network)

- Guest1 – 192.168.2.21
- Guest2 – 192.168.2.22

Each PC was configured with the correct subnet mask and default gateway.

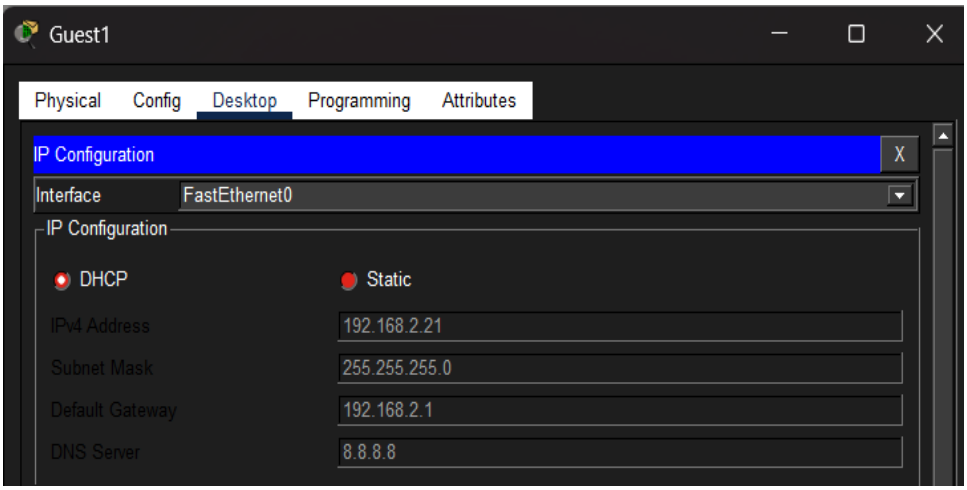

Figure 4.2: PC0 Network Address Configuration

Figure 4.2 shows the IP address configuration of the PC0 connected to both LAN1 and LAN2 networks. Each system is assigned the appropriate IP address, subnet mask, and default gateway for proper communication

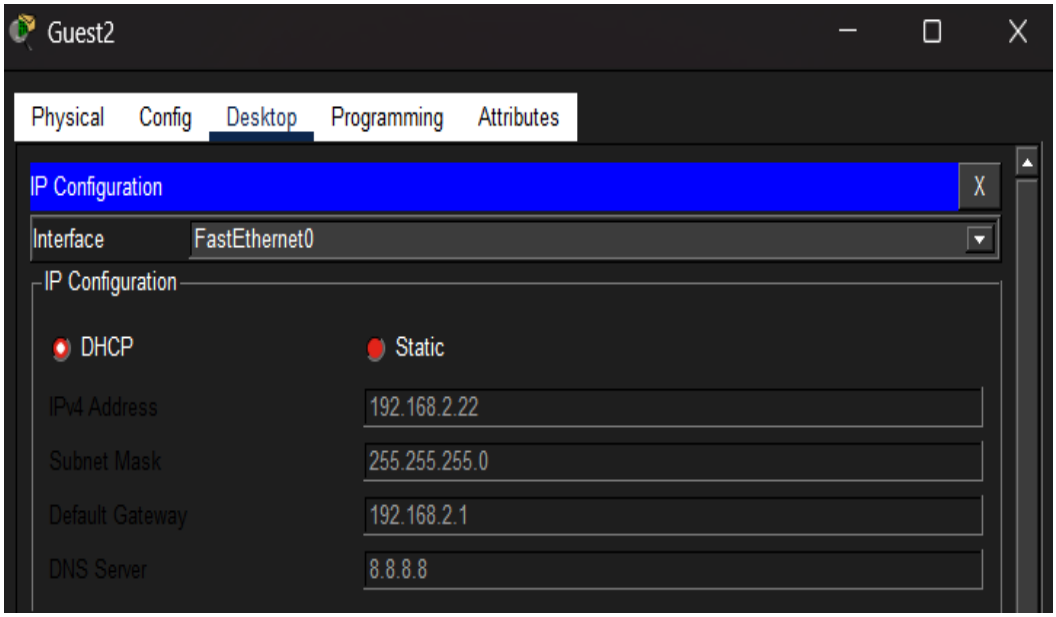

Figure 4.3: PC1 Network Address Configuration

Figure 4.3 shows the IP address configuration of the PC1 connected to both LAN1 and LAN2 networks. Each system is assigned the appropriate IP address, subnet mask, and default gateway for proper communication.

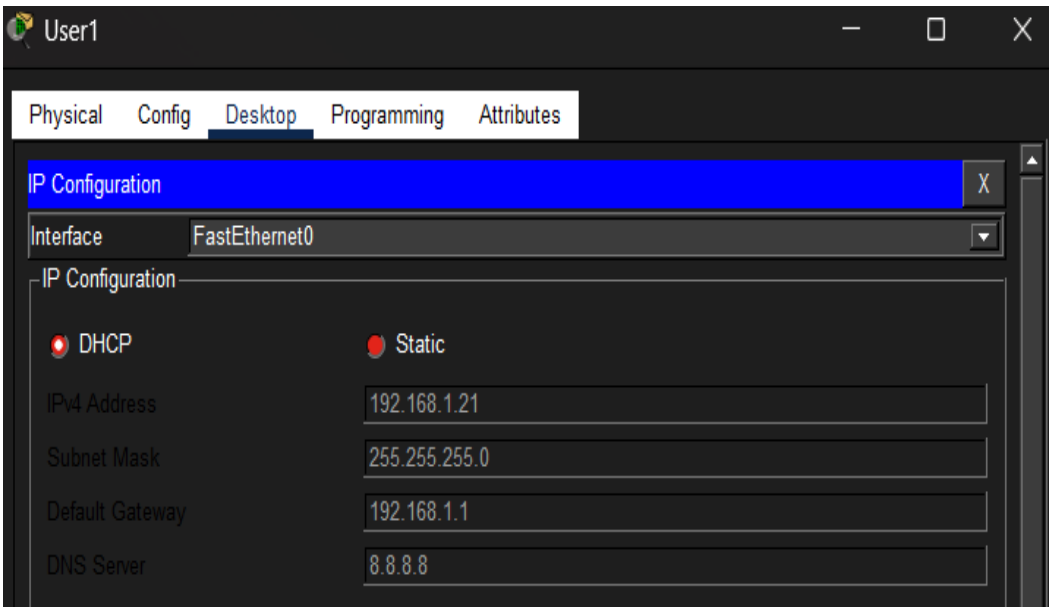

Figure 4.4: PC2 Network Address Configuration

Figure 4.4 shows the IP address configuration of the PC2 connected to both LAN1 and LAN2 networks. Each system is assigned the appropriate IP address, subnet mask, and default gateway for proper communication.

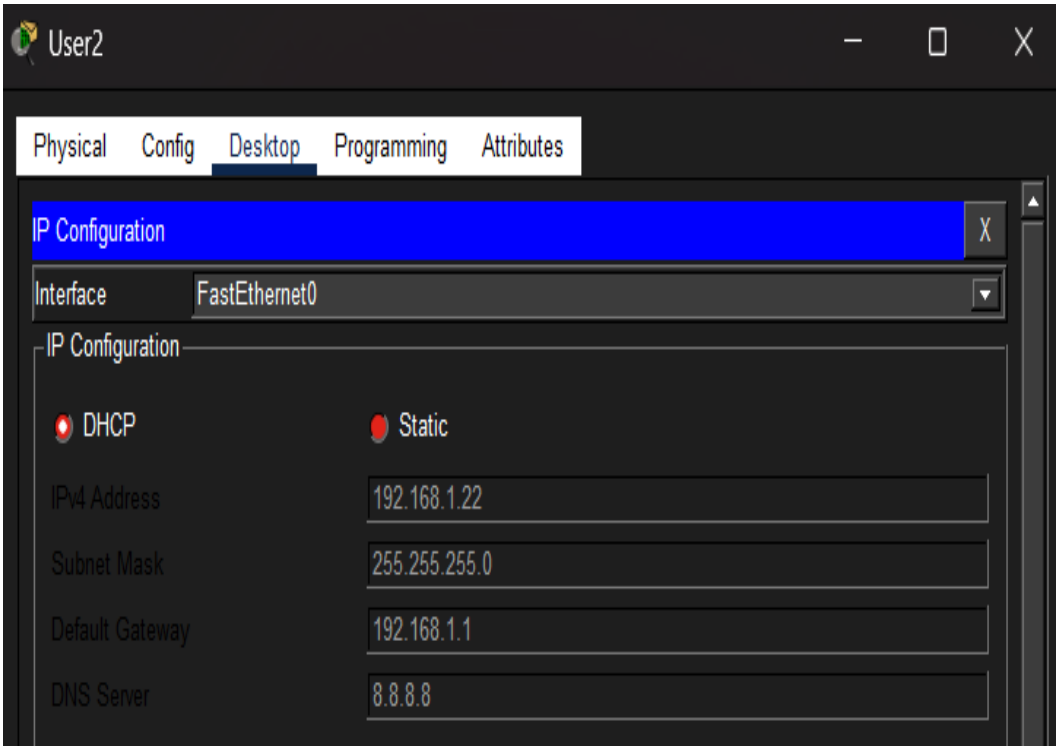

Figure 4.5: PC3 Network Address Configuration

Figure 4.5 shows the IP address configuration of the PC3 connected to both LAN1 and LAN2 networks. Each system is assigned the appropriate IP address, subnet mask, and default gateway for proper communication.

4.4 Server Configuration (HTTP Host)

The server was configured in LAN2 with the following network settings:

- IP Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

The HTTP service was enabled in Packet Tracer to host an internal webpage accessible from network.

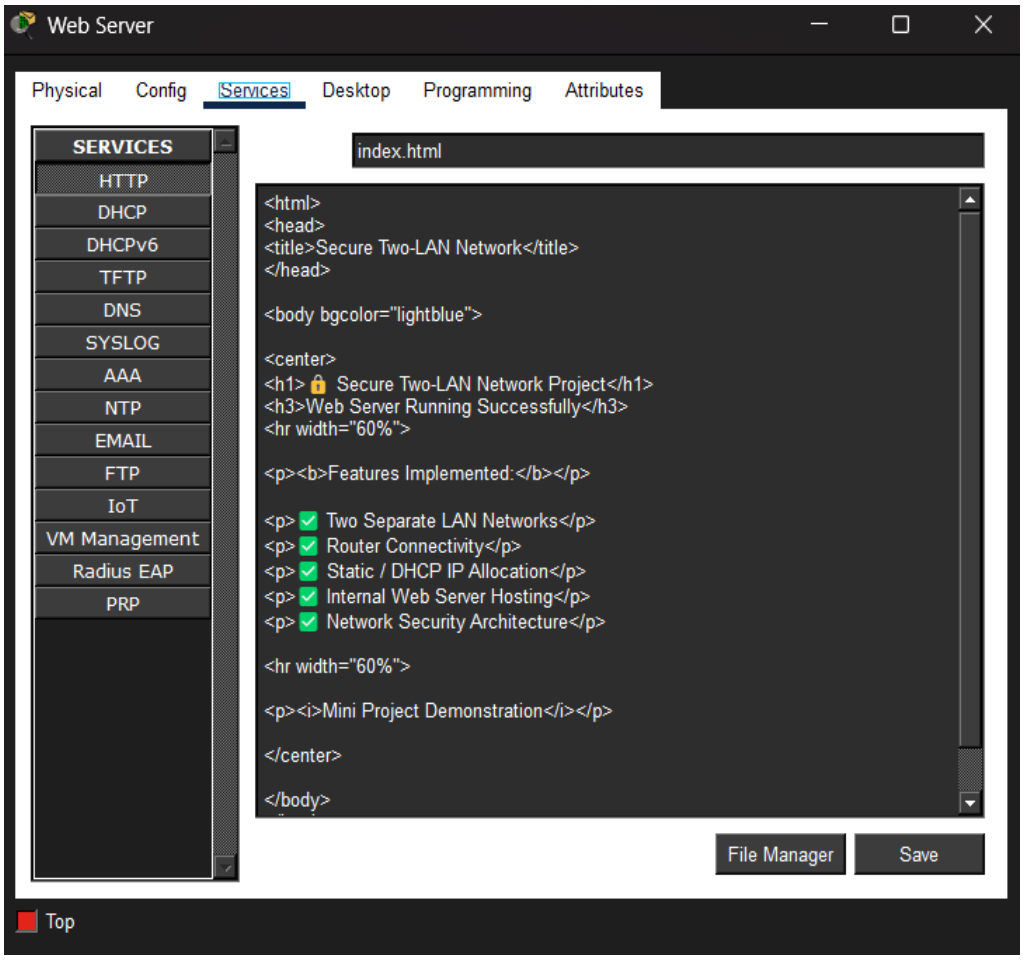

Figure 4.6: Web Server HTTP Service Configuration

Figure 4.7 shows the network configuration of the web server along with the hosted internal webpage.

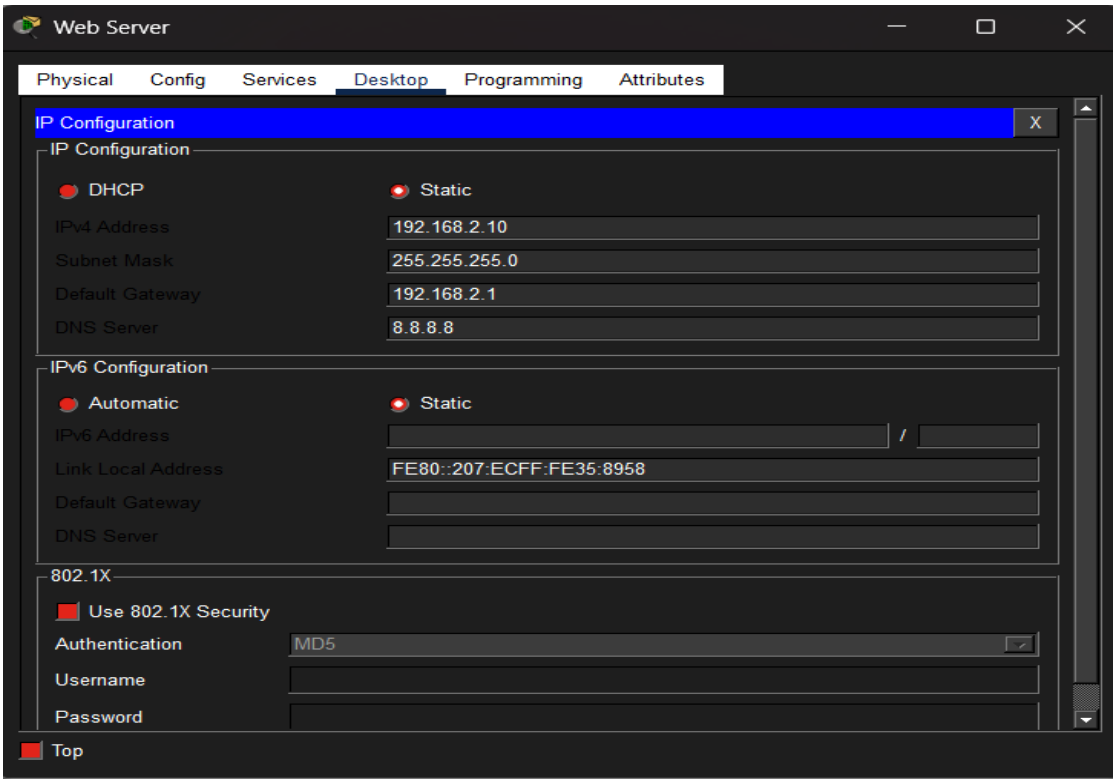

Figure 4.7: Web Server IP and Hosted Web Page Configuration

Figure 4.7 shows the network configuration of the web server along with the hosted internal webpage. It confirms proper IP settings and successful webpage hosting within the network.

4.5 Web Page and Security Configuration

A custom webpage was created on the server to verify web hosting functionality. The webpage displays project details and confirms successful deployment. Basic network security was achieved by separating devices into two LANs and controlling traffic through the router.

4.6 Ubuntu Server Setup and Apache Installation

An Ubuntu virtual machine was used to implement the practical server-side environment. Apache2 web server package was installed and started successfully. This allowed understanding of real-world server deployment beyond simulation.

```
swasthikunder@swasthikunder:~/Desktop/secure-two-lan-network$ sudo ./setup.sh
[sudo] password for swasthikunder:

🔒 Secure Two-LAN Network Setup Script v1.0.0
   Dual-LAN Segmentation + Security Hardening

✅ OS compatibility check passed
✅ All core dependencies found
✅ Log directories created at /var/log/traffic

Starting setup at Wed Apr 22 09:07:01 PM IST 2026

▶ STEP: System Update & Package Installation

ℹ️ Updating package lists...
ℹ️ Installing required packages...
```

Figure 4.8: **Ubuntu Network Setup Script Execution and Package Installation**

Figure 4.8 shows the execution of the Ubuntu network setup script used to initialize the Secure Two-LAN environment. It verifies dependency checks, log directory creation, and package installation for system configuration.

```
swasthikunder@swasthikunder:~/Desktop/secure-two-lan-network$ systemctl status apache2
Warning: The unit file, source configuration file or drop-ins of apache2.service changed on disk. Run
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-04-22 20:15:06 IST; 53min ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1403 (apache2)
      Tasks: 55 (limit: 4597)
     Memory: 7.4M (peak: 8.0M)
        CPU: 350ms
     CGroup: /system.slice/apache2.service
             ├─1403 /usr/sbin/apache2 -k start
             ├─1406 /usr/sbin/apache2 -k start
             └─1407 /usr/sbin/apache2 -k start

Apr 22 20:15:05 swasthikunder systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 22 20:15:06 swasthikunder apachectl[1387]: AH00558: apache2: Could not reliably determine the serv
Apr 22 20:15:06 swasthikunder systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)
```

Figure 4.9: Ubuntu Apache Server Installation and Status Verification

Figure 4.9 shows the installation and configuration of the Apache2 web server on the Ubuntu system. It also verifies the active server status, confirming successful deployment of the web service.

4.7 Traffic Monitoring and Log Capture

Traffic monitoring scripts were executed to capture LAN traffic and store packet capture files for analysis. Log files were generated inside the system for future inspection.

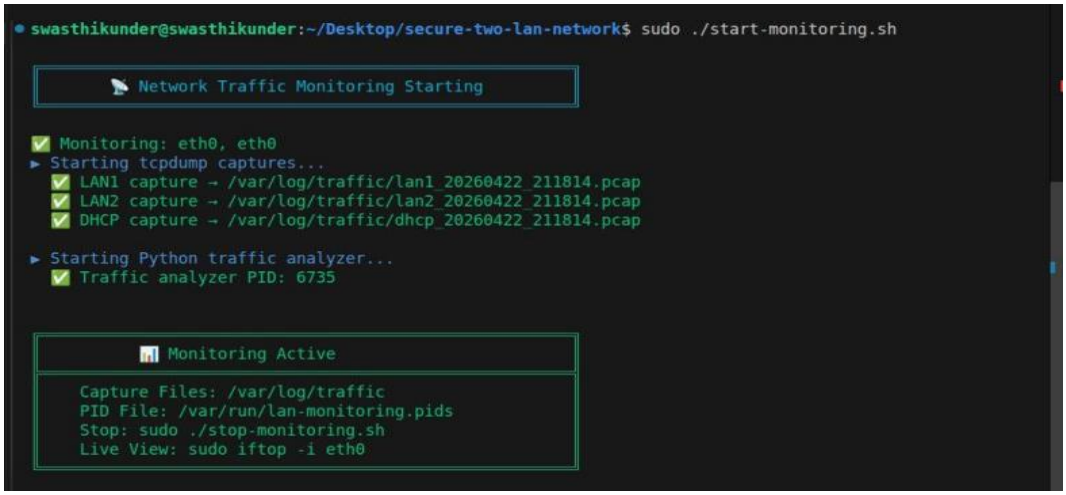

Figure 4.10: Start Monitoring Network

Figure 4.10 shows the process of stopping the active network traffic monitoring service after packet capture completion. It confirms that monitoring tasks were terminated safely and capture files were saved successfully.

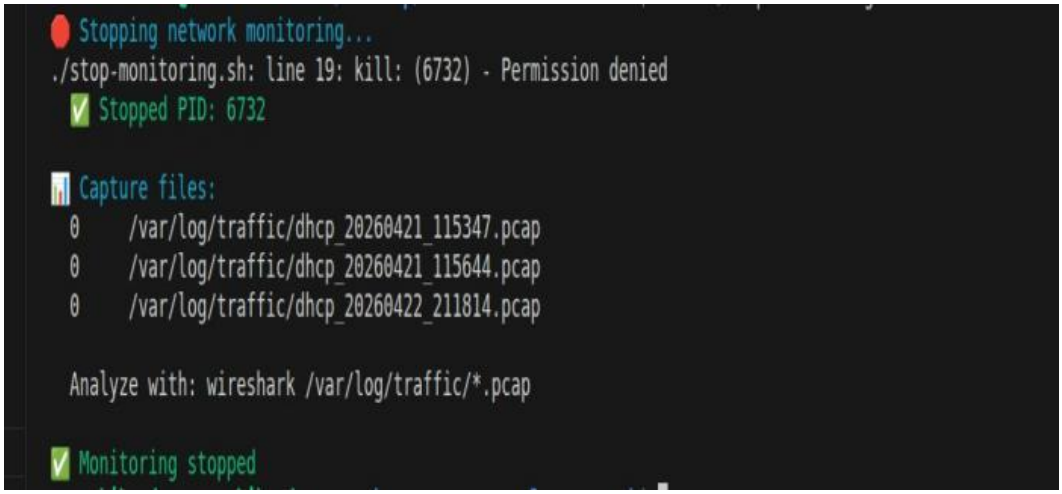

Figure 4.11: Stop Monitoring Network

Figure 4.11 shows the termination of the network traffic monitoring process after the required data was captured. It confirms successful stopping of monitoring services and proper storage of generated log files.

```
swasthikunder@swasthikunder:~/Desktop/secure-two-lan-network$ ls -lh /var/log/traffic
total 0
-rw-r--r-- 1 root    root    0 Apr 21 11:53 analyzer_20260421_115347.log
-rw-r--r-- 1 root    root    0 Apr 21 11:56 analyzer_20260421_115644.log
-rw-r--r-- 1 root    root    0 Apr 22 21:18 analyzer_20260422_211814.log
-rw-r--r-- 1 tcpdump tcpdump 0 Apr 21 11:53 dhcp_20260421_115347.pcap
-rw-r--r-- 1 tcpdump tcpdump 0 Apr 21 11:56 dhcp_20260421_115644.pcap
-rw-r--r-- 1 tcpdump tcpdump 0 Apr 22 21:18 dhcp_20260422_211814.pcap
```

Figure 4.12: Listing of Log File Generation

Figure 4.12 shows the generated log and capture files stored after executing the network monitoring process. It confirms successful recording of traffic data for further analysis and verification.

4.8 Web Hosting Verification

The localhost page was accessed through the browser to verify that the Apache web server was active and reachable

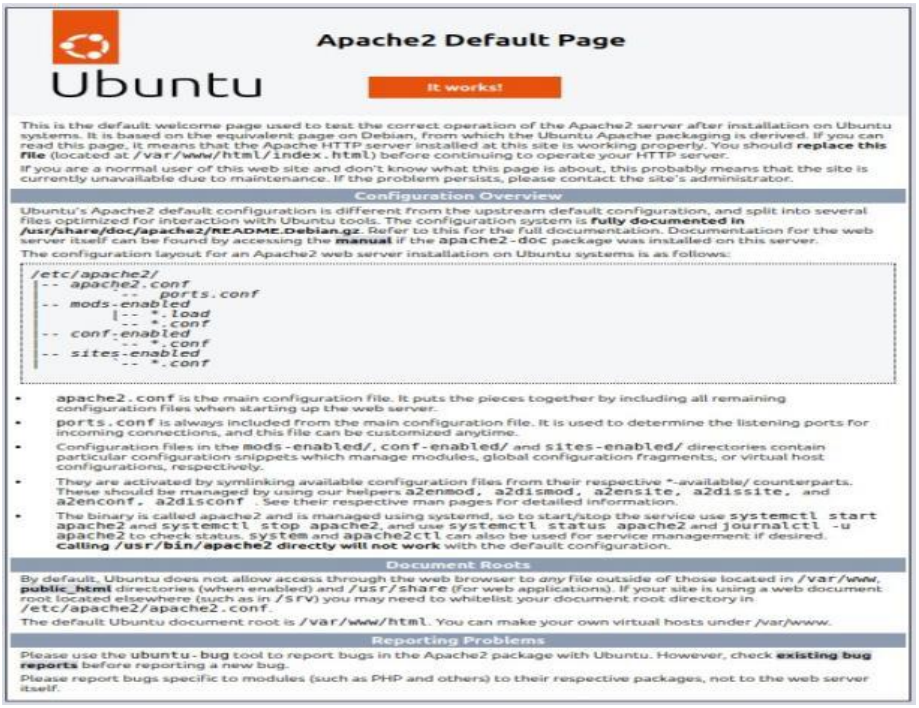

Figure 4.13: Ubuntu Apache Default Web Server Page

Figure 4.9 shows the default Apache2 web page displayed after successful installation of the web server on Ubuntu. It confirms that the Apache service is active and the server is ready to host web content.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Testing Environment

After completing the network configuration, router setup, IP addressing, and server deployment, various test cases were carried out to validate the design. These tests were performed in Cisco Packet Tracer to verify proper communication and service availability.

The following tests were conducted:

- Ping tests for verifying connectivity between devices in different LANs
- Router communication tests between LAN1 and LAN2
- Web page access to test internal server reachability
- Static and DHCP IP addressing verification
- Simulation mode testing for packet flow analysis

5.2 Ping Test Results

From	To	Result	Note
User1	User2	Success	Same LAN (LAN1)
User1	Router (192.168.1.1)	Success	Default Gateway
User1	Server(192.168.2.10)	Success	Routed LAN1 → LAN2
Guest1	Guest2	Success	Same LAN (LAN2)
Guest1	Router(192.168.2.1)	Success	Default Gateway
Guest1	User1	Success	Routed LAN2 → LAN1
User2	Server	Success	Web Server Reachable
From	**To**	**Result**	**Note**

Table 5.1 Ping test table

5.3 Website Hosting Test

The hosted web server was tested by accessing the server IP address through the web browser from user systems. The webpage loaded successfully, confirming correct

server configuration and successful communication between both LAN networks. The website was accessed using: http://192.168.2.10. This confirms that HTTP service is active and reachable through the configured network.

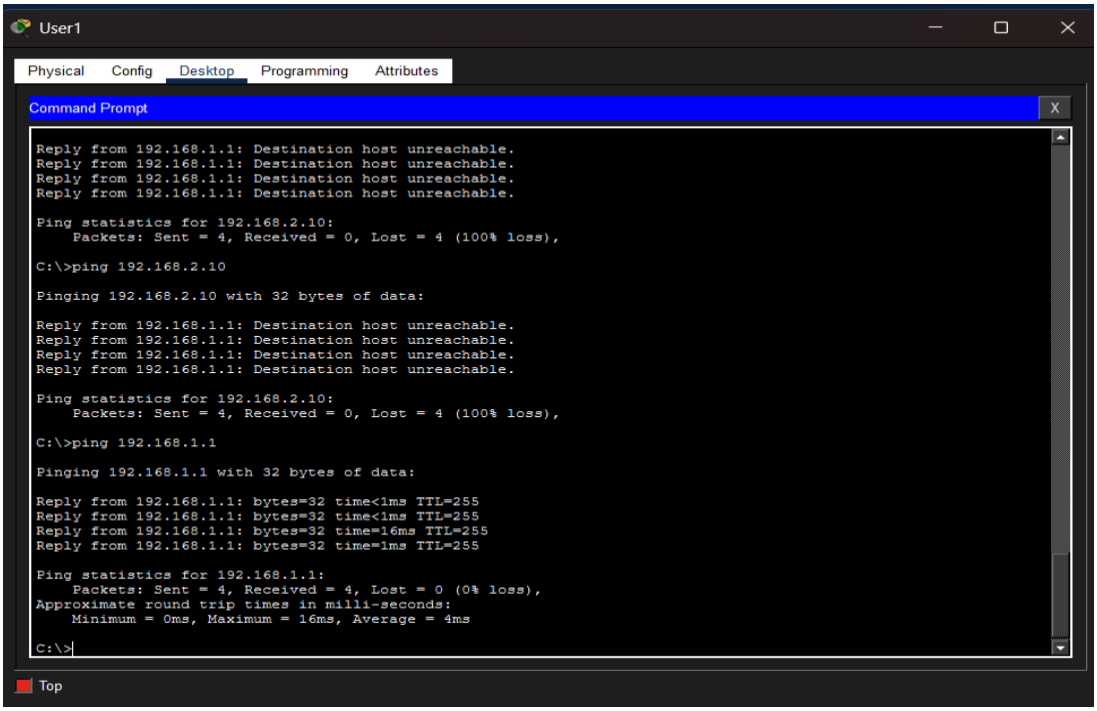

Figure 5.1: Successful Ping Test

Figure 5.2 shows the successful ping test performed between devices in different LAN networks. It confirms proper connectivity, routing, and communication across the configured network.

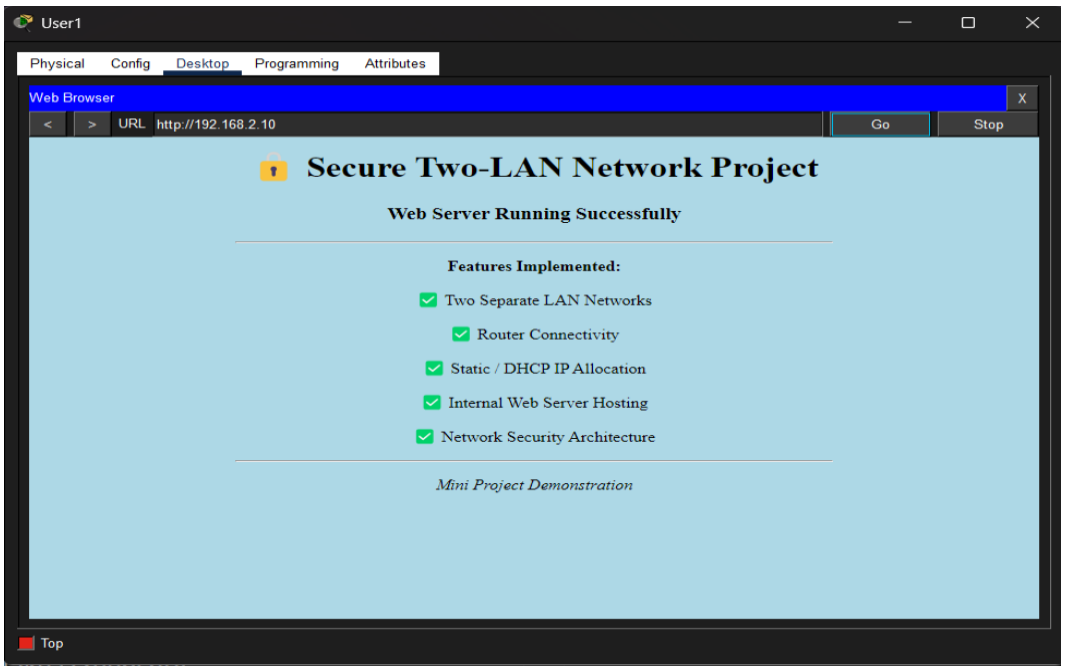

Figure 5.2: Successful Internal Website Hosting Test

Figure 5.3 shows the successful hosting of the internal webpage accessed through the web browser using the server IP address. It confirms that the HTTP service is active and reachable across the network.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project successfully demonstrates the design, configuration, and testing of a Secure Two-LAN Network using Cisco Packet Tracer and Ubuntu implementation tools. By dividing the network into two separate LANs and connecting them through a router, better traffic organization, controlled communication, and improved security were achieved.

The project successfully implemented:

• Network segmentation using two separate LANs
• Communication between LAN1 and LAN2 through router interfaces
• Static and DHCP-based IP addressing
• Internal website hosting through a web server
• Packet flow monitoring and connectivity testing
• Practical Apache server setup and traffic log generation in Ubuntu

All configured devices were able to communicate according to the network design, proving the functionality and stability of the system. The project reflects a practical network model suitable for small offices, institutions, and enterprise environments.

6.2 Future Scope

To further enhance the project and make it closer to real-world enterprise standards, the following features can be added:

• Access Control Lists (ACLs) for traffic filtering between LANs
• Firewall implementation for advanced security policies
• Internet connectivity with NAT configuration
• DHCP server for fully automatic IP allocation
• Intrusion Detection / Monitoring tools
• Wi-Fi integration for wireless clients
• Centralized logging and network monitoring server
• Real hardware implementation using Cisco devices and Ubuntu server

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   (Used for traffic capture and log monitoring concepts.)